

Managed disaster recovery and managed reader farm with Amazon RDS for Oracle using Oracle Active Data Guard

by Sameer Malik and Nagesh Battula | on 03 JUN 2020 | in [Amazon RDS](#), [RDS for Oracle](#) | [Permalink](#) | 

[Comments](#) | [Share](#)

Many AWS users are taking advantage of the managed database offerings in the [Amazon Relational Database Service](#) (Amazon RDS) portfolio to remove much of the undifferentiated heavy lifting from their day-to-day activities. With [Amazon RDS for Oracle](#), users can significantly reduce the administrative overhead of managing and maintaining an Oracle database.

Amazon RDS for Oracle provides [Multi-AZ](#) deployment to provide enhanced availability and durability for database (DB) instances within a specific AWS Region, and is often an effective disaster recovery (DR) solution for most use cases. However, several customers running mission critical databases have a business requirement for their DR configuration to span across different AWS regions. At the same time, these customers want to be able to leverage their DR investments to handle some of their production read workloads.

Currently, you can implement self-managed DR solutions for Amazon RDS for Oracle using any of the following:

- Implementing low-cost, cross-region DR for Amazon RDS for Oracle using DB snapshots. For more information, see [Cross-Region Automatic Disaster Recovery on Amazon RDS for Oracle Database Using DB Snapshots and AWS Lambda](#).
- Implementing DR for Amazon RDS for Oracle DR using [AWS Database Migration Service](#) (AWS DMS). For more information, see [Disaster recovery on Amazon RDS for Oracle using AWS DMS](#).
- Deploying cross region DR using Oracle GoldenGate for Amazon RDS for Oracle. For more information, see [Implementing cross-region disaster recovery using Oracle GoldenGate for Amazon RDS for Oracle](#).

With in-region and cross-region Read Replicas support on Amazon RDS for Oracle, you can easily create the replicas using Oracle Active Data Guard. This post discusses the managed DR and managed reader farm capabilities of Amazon RDS for Oracle Read Replicas. It also covers other use cases such as offloading read workloads from the primary DB instance, scaling out read workloads over a reader farm, and data proximity.

Introduction to Amazon RDS for Oracle Read Replicas

Amazon RDS for Oracle Read Replicas (both in-region and cross-region) use Oracle Active Data Guard between the primary and the physical standby (Read Replica) DB instances. Amazon RDS for Oracle configures asynchronous Data Guard replication between the primary and the Read Replica, where the Read Replica is open in read-only mode while the Redo Apply is active. The primary and the Read Replica DB instances operate in Data Guard Max Performance mode, which uses log writer (LGWR) asynchronous transport (ASYNC), with no affirmation on the Read Replica I/O (NOAFFIRM). In this mode, the transactions are committed as soon as the redo generated is written to the online redo log. Additionally, the redo is written to the Read Replica's redo logs asynchronously. This solution provides the highest level of data protection possible without impacting the performance of the primary DB instance.

Depending on the database workload and amount of redo logs it generates, you can achieve low RPO and RTO in single-digit minutes. RDS Oracle creates a secure communication channel between the primary and Read

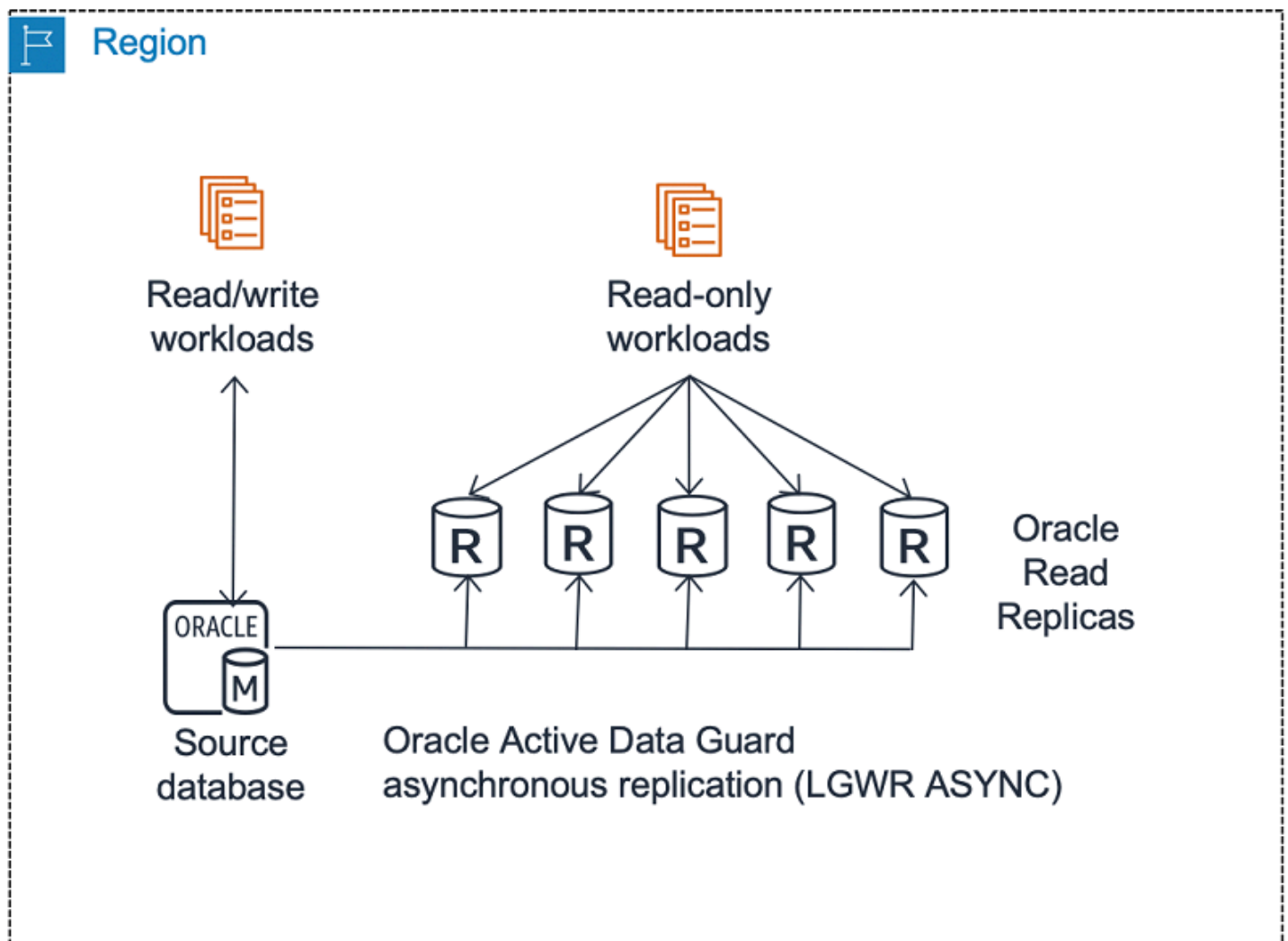
Replicas, and manages the setup and configuration of security groups and access control lists to enable secure redo transport between the primary and Read Replicas.

Use cases for In-region Read Replicas

The following are possible use cases for in-region Read Replicas:

- **Offload read workloads** – Customers can offload their read workload from the primary DB instance to the Read Replica DB instance and therefore boost the performance of read-write transactions running on the primary
- **Scale out read workloads** – Customers can scale out read workloads over a reader farm of up to five Read Replicas
- **Availability** – Customers can promote a given Read Replica as a new standalone database to achieve additional availability in case of primary DB failure

The following diagram illustrates the architecture for Amazon RDS for Oracle In-region Read Replicas.



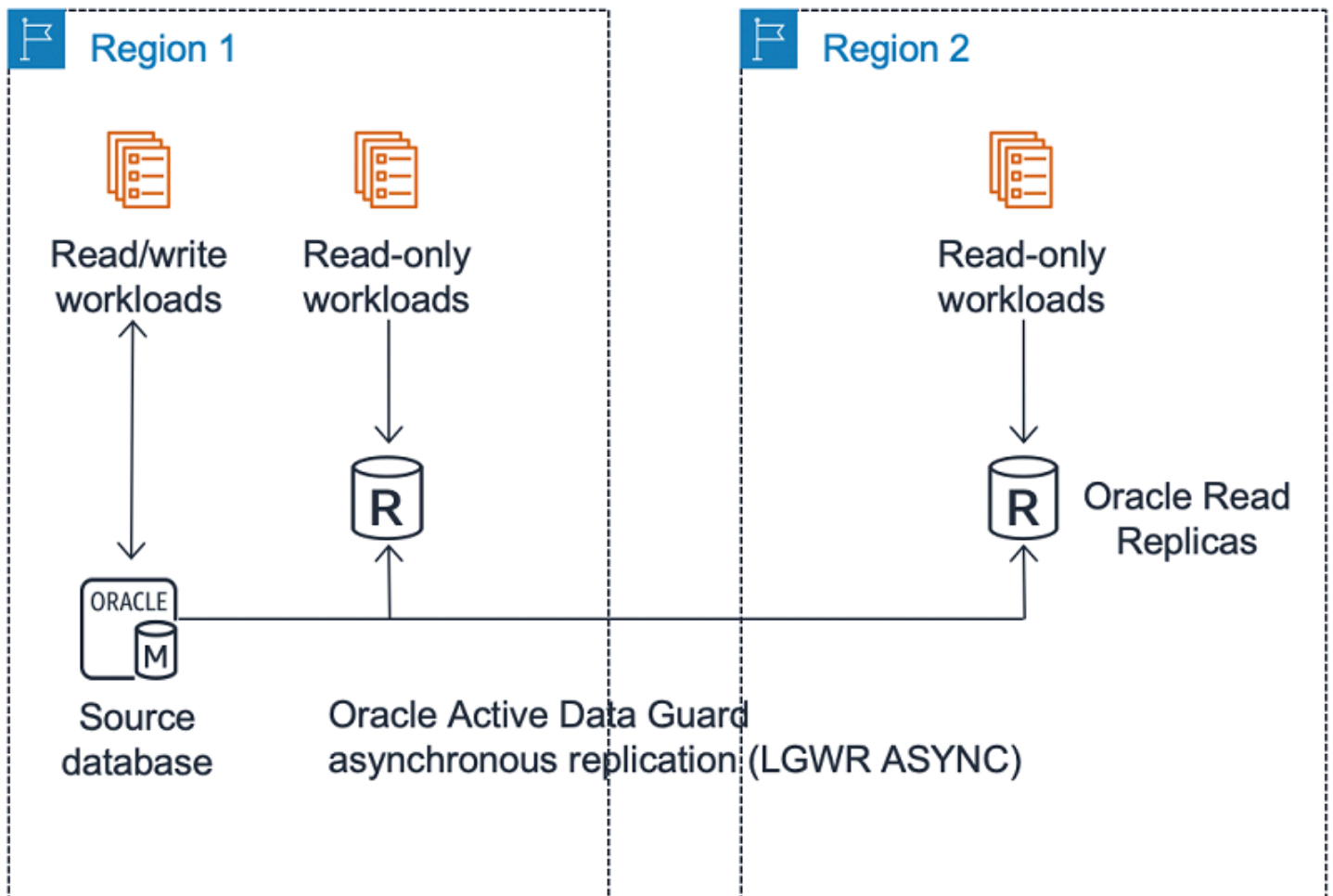
Use cases for Cross-region Read Replicas

The following are possible use cases for cross-region Read Replicas:

- **Managed disaster recovery** – Using managed DR offering customers can promote a Read Replica in another Region to be a new standalone production database

- **Data proximity** – Customers can place replicas closer to the application users of a given Region to reduce read latencies
- **Offload read workloads** – Customers can offload their read workload from the primary DB instance to the Read Replica DB instance . The Read Replicas can reside in any AWS Region.
- **Scale out read workloads** – Customers can scale out read workloads over a reader farm of up to five Read Replicas that can reside in any AWS Region

The following diagram illustrates the architecture for Amazon RDS for Oracle Cross-region Read Replicas.



Creating Read Replicas on the Amazon RDS console

Amazon RDS for Oracle Read Replicas provide enhanced performance and durability for DB instances. Amazon RDS creates a second DB instance using a snapshot of the source DB instance. To enable Read Replicas, you must enable automatic backups (set the backup retention period to a value greater than 0) and force logging on the source DB instance. To enable force logging mode, connect to the DB instance and enter the following code:

```
exec rdsadmin.rdsadmin_util.force_logging(p_enable => true);
```

For more information about creating Read Replicas and their limitations, see [Read Replica Limitations with Oracle](#).

This section walks you through the creation of cross-region Read Replica.

1. Create Primary (source DB) instance.

For demonstrating the creation of Read Replica, the primary Database instance “mydbinstance” is already created in the AWS region “us-west-2” using the below CLI command.

```
aws rds create-db-instance \
  --db-instance-identifier mydbinstance \
  --db-name MYDB \
  --allocated-storage 20 \
  --storage-type gp2 \
  --db-instance-class db.m5.xlarge \
  --engine oracle-ee \
  --port 1521 \
  --backup-retention-period 1 \
  --license-model byol \
  --master-user-password xxxxxxxxxxxx \
  --master-username admin \
  --engine-version 19.0.0.0.ru-2020-01.rur-2020-01.r1
```

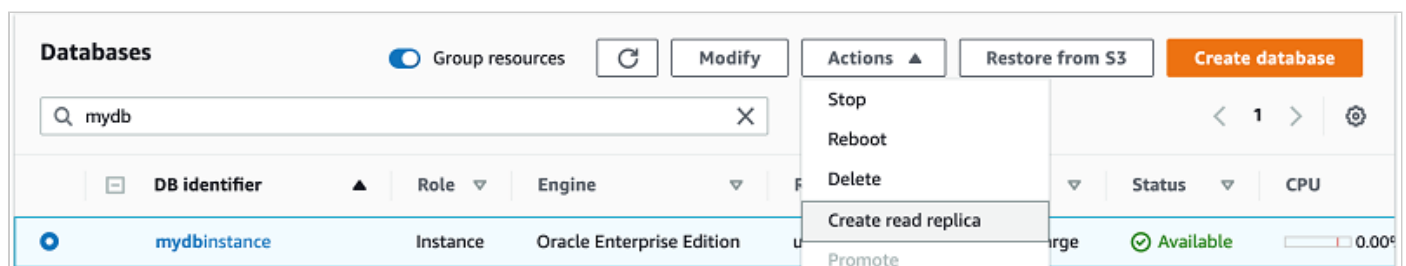
For more information, see [Creating an Oracle DB Instance and Connecting to a Database on an Oracle DB Instance](#).

When the primary DB instance status shows as **Available**, you can create the Read Replica (make sure to switch the Amazon RDS console to the Region where the primary DB instance is located).

2. On the Amazon RDS console, choose **Databases**.

3. Select **mydbinstance**.

4. From the **Actions** drop-down menu, choose **Create read replica**.



5. In the **Instance specifications** section, for **DB instance class**, choose **m5.xlarge**.

6. For **Multi-AZ deployment**, select **No**.

7. For **Storage type**, choose **General Purpose (SSD)**.

[RDS](#) > [Databases](#) > Add reader

Create read replica DB instance

You are creating a replica DB instance from a source DB instance. This new DB instance will have the source DB instance's DB security groups and DB parameter groups.

Instance specifications

DB instance class

Contains the compute and memory capacity of the DB instance.

db.m5.xlarge

Multi-AZ deployment

Specifies if the DB instance should have a standby deployed in another availability zone.

☐ Yes

☒ No

Storage type [Info](#)

General Purpose (SSD)

8. In the **Network & Security** section, for **Destination region**, choose **US East (N. Virginia)**.
9. For **Destination DB subnet group**, choose **default**.
10. For **Availability zone**, choose **No preference**.
11. For **Publicly accessible**, select **Yes**.
12. For **VPC security groups**, choose **default**.

Network & Security

Destination region
The region in which the replica will be launched

US East (N. Virginia) ▼

Destination DB subnet group

default ▼

Availability zone
The EC2 Availability Zone that the database instance will be created in.

No preference ▼

Publicly accessible

☒ **Yes**
EC2 instances and devices outside of the VPC hosting the DB instance will connect to the DB instances. You must also select one or more VPC security groups that specify which EC2 instances and devices can connect to the DB instance.

☐ **No**
DB instance will not have a public IP address assigned. No EC2 instance or devices outside of the VPC will be able to connect.

VPC security groups

Choose VPC security groups ▼

default X

13. In the **Settings** section, for **Read replica source**, choose **mydbinstance**.

14. For **DB instance identifier**, enter `mydbinstance-ro`.

Encryption

Encryption

☒ Enable encryption [Learn more](#) 

Select to encrypt the given instance. Master key ids and aliases appear in the list after they have been created using the Key Management Service(KMS) console.

☐ Disable encryption

Master key [Info](#)

(default) aws/rds ▼

Account

KMS key ID

Settings

Read replica source

Source DB instance Identifier

mydbinstance ▼

DB instance identifier

DB instance identifier. This is the unique key that identifies a DB instance. This parameter is stored as a lowercase string (e.g. mydbinstance).

mydbinstance-ro

15. In the **Monitoring** section, for **Enhanced monitoring**, select **Disable enhanced monitoring**.

16. In the **Log exports** section, leave all log types deselected.

Monitoring

Enhanced monitoring

☒ **Enable enhanced monitoring**
Enhanced monitoring metrics are useful when you want to see how different processes or threads use the CPU.

☐ **Disable enhanced monitoring**

Monitoring Role Granularity

Default 5 seconds

☒ I authorize RDS to create the IAM role rds-monitoring-role.

Log exports

Select the log types to publish to Amazon CloudWatch Logs

☐ Alert log

☐ Audit log

☐ Listener log

☐ Trace log

IAM role
The following service-linked role is used for publishing logs to CloudWatch Logs.

RDS Service Linked Role

17. In the **Performance Insights** section, select **Disable Performance Insights**.

18. In the **Maintenance** section, for **Auto minor version upgrade**, select **No**.

Performance Insights

☒ Enable Performance Insights
☐ Disable Performance Insights

Retention period [Info](#)

Default (7 days) ▼

Master key [Info](#)

(default) aws/rds ▼

⚠ Attention: the KMS key cannot be changed after enabling Performance Insights

Maintenance

Auto minor version upgrade
 Specifies if the DB instance should receive automatic engine version upgrades when they are available.

☐ Yes
☒ No

[Cancel](#) [Create read replica](#)

19. Choose **Create read replica**.

Creation takes 10–15 minutes to complete, after which the role for the primary DB instance shows as

Master in **us-west-2**.

Databases						
☒ Group resources			Modify	Actions ▼	Restore from S3	Create
mydb						
☐ DB identifier	Role ▼	Engine	Region & AZ ▼	Size	Status ▼	
☒ mydbinstance	Master	Oracle Enterprise Edition	us-west-2b	db.m5.xlarge	✓ Available	

The role for the Read Replica shows as **Replica** in **us-east-1**.

Databases						
☒ Group resources			Modify	Actions ▼	Restore from S3	Create database
mydb						
☐ DB identifier	Role ▼	Engine	Region & AZ ▼	Size	Status	
☐ mydbinstance-ro	Replica	Oracle Enterprise Edition	us-east-1a	db.m5.xlarge	✓ Avail	

Creating the Read Replica in the AWS CLI

Alternatively, you can also create the Read Replica in the AWS CLI or with the [create-db-instance-read-replica](#) Amazon RDS API. To create the cross-region Read Replica (**mydbinstance-ro**) in the **us-east-1** with the

primary DB instance (`mydbinstance`) running in `us-west-2` , enter the following code:

```
aws rds create-db-instance-read-replica \
--source-db-instance-identifier arn:aws:rds:us-west-2:xxxxxxx:db:mydbinstance\
--source-region us-west-2\
--db-instance-identifier mydbinstance-ro\
--region us-east-1
-- kms-key-id kms-key-id xxxxxxxx: xxxxxxxx:
```

If you create an encrypted Read Replica in a different Region, you must specify an AWS KMS key for the destination Region. AWS KMS encryption keys are specific to the Region that you create them in; you can't use encryption keys from one Region in another Region.

Connecting to the Read Replica

You can connect to the Read Replica instance using your preferred client; this post uses SQL* Plus as shown below. For more information about connecting to the RDS Oracle database, see [Connecting to Your Sample Oracle DB Instance](#).

To connect to the Read Replica using SQL* PLUS, enter the following code:

```
$ sqlplus 'admin@(DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST= mydbinstance-ro.xxxxxx

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
With the Partitioning, OLAP, Advanced Analytics and Real Application Testing optio

SQL> set lines 250 pages 10000
SQL> show parameter db_uniq

NAME                                TYPE      VALUE
-----
db_unique_name                      string    MYDB_B

SQL> select open_mode,database_role from v$database;

OPEN_MODE          DATABASE_ROLE
-----
READ ONLY WITH APPLY    PHYSICAL STANDBY
```

Monitoring a Read Replica lag

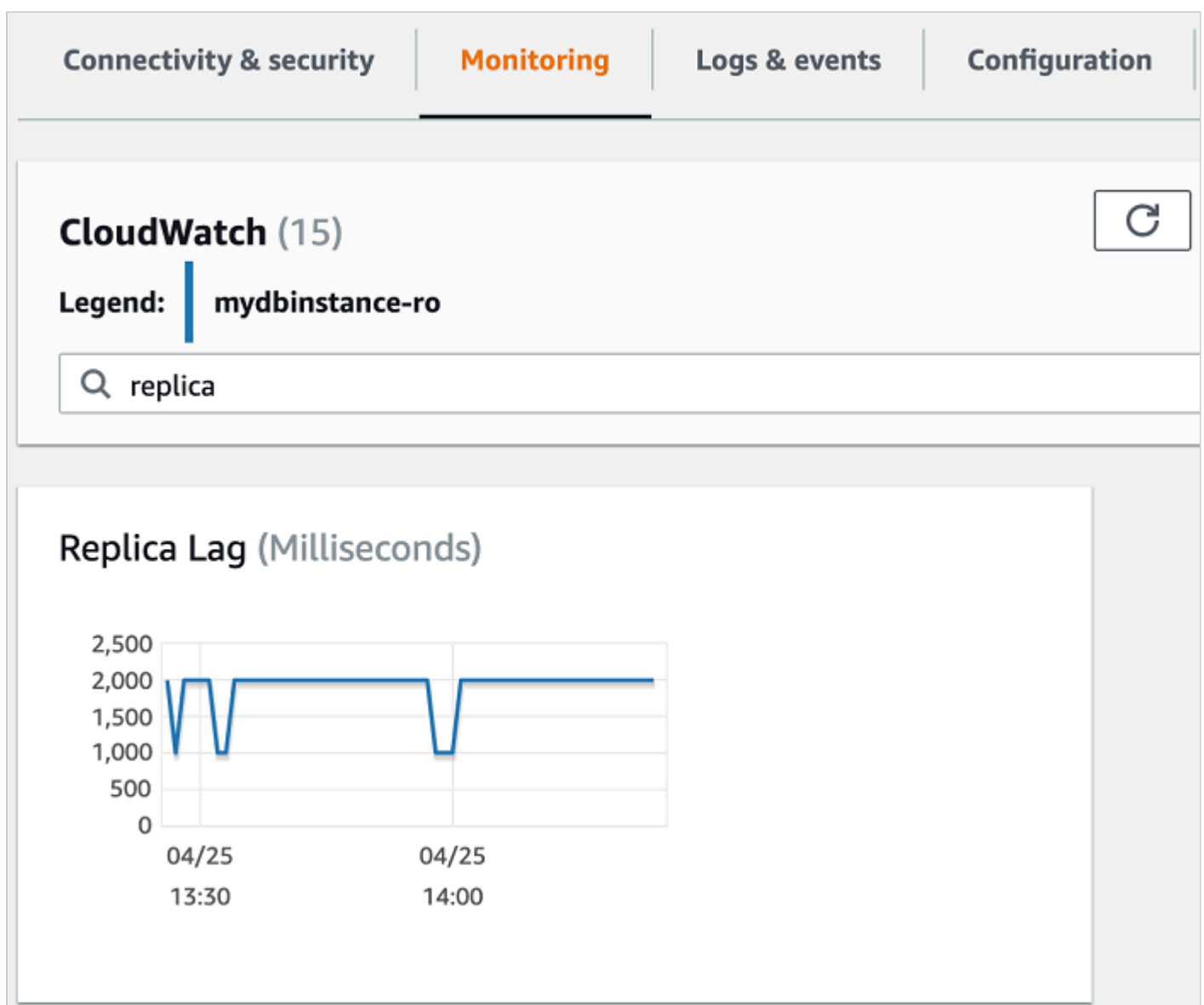
You can monitor the replication lag on [Amazon CloudWatch](#) console, the Amazon RDS console, Oracle Enterprise Manager (OEM), or with Oracle Data Guard views (`v$archived_log` , `v$dataguard_stats` ,

```
v$dataguard_status ).
```

To monitor on the Amazon RDS console, on the **Databases** page, select the Read Replica instance.

Databases						
Group resources Refresh Modify Actions Restore from S3 Create database						
mydb < 1 > Settings						
DB identifier	Role	Engine	Region & AZ	Size	Status	
mydbinstance-ro	Replica	Oracle Enterprise Edition	us-east-1a	db.m5.xlarge	Available	

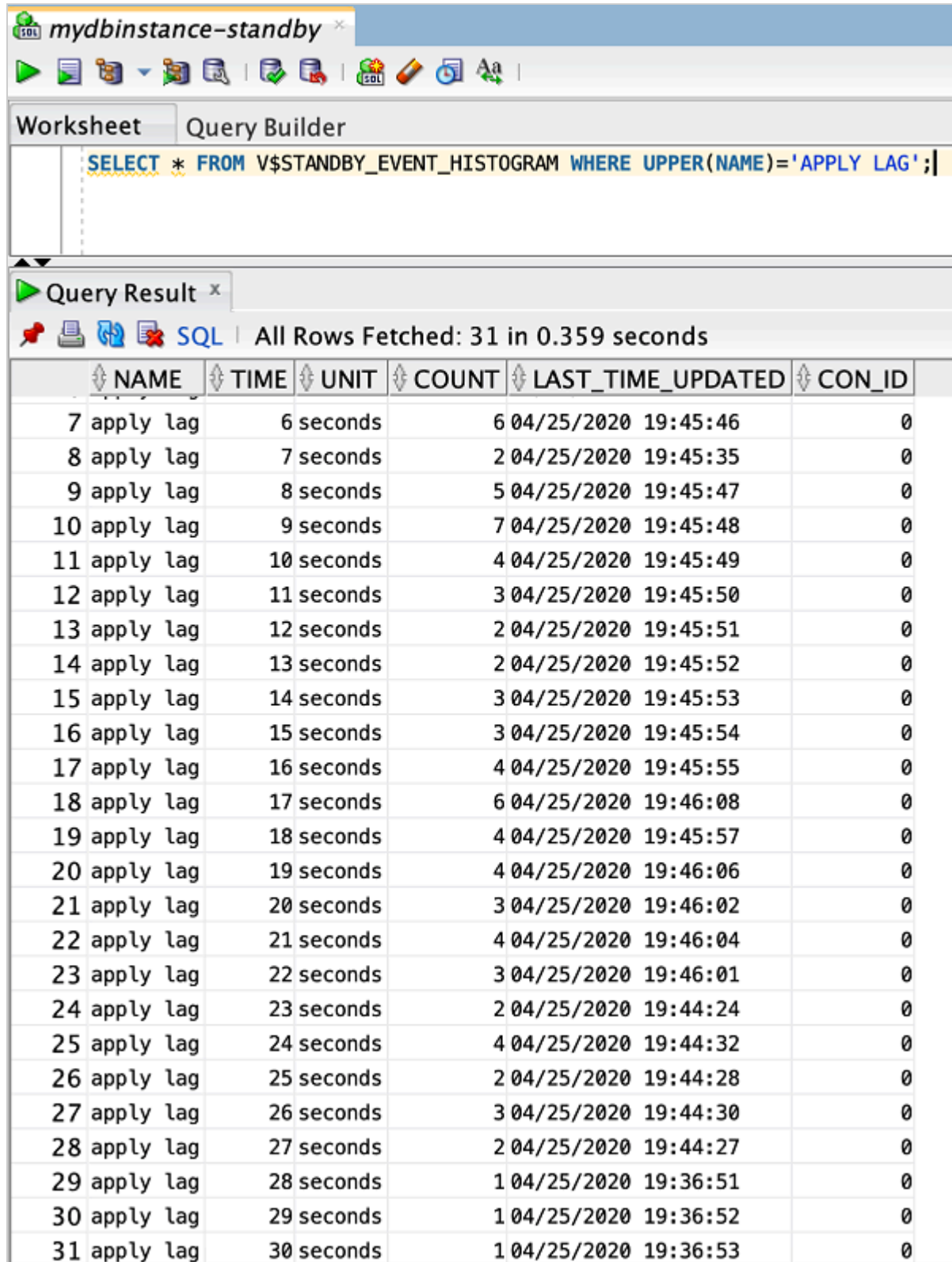
On the **Monitoring** tab, search for **replica lag**.



You can connect to the Read Replica instance using your preferred client; this post uses SQL Developer. For more information about connecting to the RDS Oracle database, see [Connecting to Your Sample Oracle DB Instance](#). See the following code:

You can run the following queries on the Read Replica instance to check more details on the replication lag using Oracle Active Data Guard views. Below is the output from the Read Replica connected as admin user using SQL Developer.

```
SQL> Select * from v$standby_event_histogram where upper(name)='APPLY_LAG';
```



The screenshot shows the SQL Developer interface with a worksheet titled 'mydbinstance-standby'. The query entered is `SELECT * FROM V$STANDBY_EVENT_HISTOGRAM WHERE UPPER(NAME)='APPLY LAG';`. The query result is displayed in a table with 7 columns: NAME, TIME, UNIT, COUNT, LAST_TIME_UPDATED, and CON_ID. The table contains 31 rows of data, showing the replication lag for the 'APPLY_LAG' event across different standby databases.

	NAME	TIME	UNIT	COUNT	LAST_TIME_UPDATED	CON_ID
7	apply lag	6 seconds		6	04/25/2020 19:45:46	0
8	apply lag	7 seconds		2	04/25/2020 19:45:35	0
9	apply lag	8 seconds		5	04/25/2020 19:45:47	0
10	apply lag	9 seconds		7	04/25/2020 19:45:48	0
11	apply lag	10 seconds		4	04/25/2020 19:45:49	0
12	apply lag	11 seconds		3	04/25/2020 19:45:50	0
13	apply lag	12 seconds		2	04/25/2020 19:45:51	0
14	apply lag	13 seconds		2	04/25/2020 19:45:52	0
15	apply lag	14 seconds		3	04/25/2020 19:45:53	0
16	apply lag	15 seconds		3	04/25/2020 19:45:54	0
17	apply lag	16 seconds		4	04/25/2020 19:45:55	0
18	apply lag	17 seconds		6	04/25/2020 19:46:08	0
19	apply lag	18 seconds		4	04/25/2020 19:45:57	0
20	apply lag	19 seconds		4	04/25/2020 19:46:06	0
21	apply lag	20 seconds		3	04/25/2020 19:46:02	0
22	apply lag	21 seconds		4	04/25/2020 19:46:04	0
23	apply lag	22 seconds		3	04/25/2020 19:46:01	0
24	apply lag	23 seconds		2	04/25/2020 19:44:24	0
25	apply lag	24 seconds		4	04/25/2020 19:44:32	0
26	apply lag	25 seconds		2	04/25/2020 19:44:28	0
27	apply lag	26 seconds		3	04/25/2020 19:44:30	0
28	apply lag	27 seconds		2	04/25/2020 19:44:27	0
29	apply lag	28 seconds		1	04/25/2020 19:36:51	0
30	apply lag	29 seconds		1	04/25/2020 19:36:52	0
31	apply lag	30 seconds		1	04/25/2020 19:36:53	0

```
SQL> Select name,value,datum_time,time_computed from v$dataguard_stats where upper(r
```

The screenshot shows the SQL Developer interface for the database instance 'mydbinstance-standby'. The 'Query Result' window displays the following data:

NAME	VALUE	DATUM_TIME	TIME_COMPUTED
1 apply lag	+00 00:00:00	04/25/2020 21:19:29	04/25/2020 21:19:31

The query executed was: `SELECT NAME,VALUE,DATUM_TIME,TIME_COMPUTED FROM V$DATAGUARD_STATS WHERE UPPER(NAME)='APPLY LAG';`

Promoting a Read Replica

In the case of a disaster, complete the following steps to promote a given cross-region Read Replica as a new standalone production database:

1. Stop the applications that are accessing the old production database.
2. Make sure all the available redo logs have been applied to the cross-region Read Replica and there is no replication lag.
3. Promote the cross-region Read Replica.
4. Point applications to the listener endpoint of the newly promoted database.
5. Recreate Read Replicas off of the newly activated standalone database.

To promote the Read Replica manually, complete the following steps:

1. On the **Databases** page, select **mydbinstance-ro**.
2. From the **Actions** drop-down menu, choose **Promote**.

The screenshot shows the Amazon RDS console 'Databases' page. The 'mydbinstance-ro' Read Replica is selected, and the 'Promote' action is highlighted in the dropdown menu. The table below shows the details of the selected instance:

DB identifier	Role	Engine	& AZ	Size	Status
mydbinstance-ro	Replica	Oracle E	-1a	db.m5.xlarge	✓ Avail

3. On the **Promote Read Replica** page, for **Enable automatic backups**, select **Yes**.
4. For **Backup retention period**, choose **1 days**.
5. For **Backup window**, select **No preference**.

RDS > Databases > Promote read replica

Promote Read Replica: mydbinstance-ro

Preferences

Enable automatic backups

☒ Yes☐ No

Selecting no will disable automated backups

Backup retention period

The number of days for which automated backups are retained. Setting this parameter to a positive number enables backups. Setting this parameter to 0 disables automated backups.

1



days

Backup window

The daily time range (in UTC) during which automated backups are created if automated backups are enabled.

☐ Select Window☒ No preference

6. Choose **Continue**.

You now see a confirmation page.

7. Choose **Promote Read Replica**.

RDS > Databases > Promote read replica

Promote Read Replica: mydbinstance-ro

Are you sure you want to promote this Read Replica?

Before you promote this Read Replica, we recommend that you stop any transactions on the master and wait for the Read Replica lag to be zero. Otherwise, there is a high likelihood that the Read Replica does not have all the transactions committed to the master DB Instance.

Note that the promotion process takes a few minutes to complete. When you promote a Read Replica, replication is stopped and the Read Replica is rebooted as part of the promotion. In addition, the promotion process is irreversible and you cannot restart the replication with the promoted DB Instance as a replication target.

Are you sure you want to promote this Read Replica?

[Cancel](#)[Back](#)[Promote Read Replica](#)

The process takes approximately 5 minutes. When it is successfully complete, you can verify the status of the master and replica instances on the Amazon RDS console. The role for both shows as **Instance**.

RDS > Databases

Databases ☒ Group resources  **Modify** **Actions** ▼

Q mydb X

DB identifier ▲	Role ▼	Engine ▼	Region & AZ ▼
☒ mydbinstance	Instance	Oracle Enterprise Edition	us-west-2b

Databases ☒ Group resources  **Modify** **Actions** ▼ **Restore from S3**

Q mydb X

DB identifier ▲	Role ▼	Engine ▼	Region & AZ ▼
☒ mydbinstance-ro	Instance	Oracle Enterprise Edition	us-east-1a

You can modify the DB instance identifier if needed by selecting the instance and choosing **Modify**.

Promoting a Read Replica via AWS CLI

To promote a Read Replica on the AWS CLI, enter the following code:

```
aws rds promote-read-replica \
  --db-instance-identifier mydbinstance-ro\
```

Planned maintenance

For both the major and minor upgrades, the engine upgrade is applied on the primary DB instance first, followed by applying the engine upgrade on all the replicas in parallel (Please note all the replicas are upgraded in the primary DB instance's maintenance window). For more information about upgrading the database engine for Amazon RDS for Oracle, see [Upgrading the Oracle DB Engine](#).

Licensing requirements

Amazon RDS for Oracle supports both in-region and cross-region Read Replica features if you're using the bring your own license (BYOL) model with Oracle Database Enterprise Edition and own a license for the Oracle Active Data Guard option. You have to license Oracle Enterprise Edition and the Oracle Active Data Guard option for the primary DB instance and each Read Replica DB instance.

Compatibility

Amazon RDS for Oracle supports this feature for Oracle Enterprise Edition versions of 12.1 (starting from 12.1.0.2.v10) and higher.

Billing

A Read replica is billed as a standard DB instance and at the same rates. Just like a standard DB instance, the DB instance class of the Read Replica determines the rate per DB instance hour.

For Cross-region Read Replicas, the data transfer between the primary DB instance and the Read Replica DB instance across Regions is billed based on the data transfer rates for the source and destination regions. For more information about instance pricing, storage, data transfer, and regional availability, see [Amazon RDS for Oracle pricing](#).

High availability for the Read Replicas

You can enable Multi-AZ to attain high availability for business-critical reporting workloads running on the Read Replicas. Upon Multi-AZ failovers (either on the primary or the Read Replica environments), the Data Guard replication between the source and the target is automatically reconfigured and no intervention is required.

The following CLI code shows how to enable Multi-AZ on the Read Replica.

```
## Modify read replica: convert SAZ to MAZ:

aws rds modify-db-instance \
  --db-instance-identifier mydbinstance-ro\
  --apply-immediately\
  -- multi-az
```

The instance class and the storage type you use for Read Replicas don't need to be identical to that of the primary. For example, the in-region Read Replica could use a lower instance class, size, and storage type—just enough to handle the read workloads. If you want to use a cross-region Read Replica for DR, you should mirror it with that of the primary's instance class, size, and the storage type so that when the cross-region Read Replica is promoted, it can handle all the production workload.

Using Multi-AZ vs. Read Replicas

Amazon RDS for Oracle Read Replicas are complimentary to Amazon RDS Multi-AZ deployments. While the Amazon RDS Multi-AZ provides high availability and data protection, the associated in region Read Replica renders the scalability of read-only workloads. While the cross region Read Replica provides Disaster Recovery in addition to the read scalability. Furthermore, if you want to run business-critical reporting workloads on the Read Replicas, you should enable Multi-AZ on the Read Replica DB instances. The following table compares Amazon RDS Multi-AZ with Amazon RDS for Oracle Read Replicas

Multi-AZ	Read Replicas
----------	---------------

Synchronous replication—highly durable with zero data loss	Asynchronous replication—highly scalable
Primary use case High Availability and Disaster recovery within a single AWS region	Primary use case for in region Read Replica is read scalability and for cross region Read Replica it is Disaster Recovery across 2 AWS regions
Only primary instance is active at any point in time	All replicas are active and available for read scaling
Backups can come from secondary	RDS Oracle Read Replicas don't support backups
Always in two Availability Zones within a Region	Can be within an Availability Zone, cross Availability Zone, or cross-region
Automatic failover when a problem is detected	Can be manually promoted to a standalone database

Summary

Amazon RDS for Oracle supports managed DR and managed scalability of read workloads over a farm of Read Replicas. If you're an Oracle BYOL customer and need a DR strategy, you can move your self-managed mission-critical databases running in your data centers or on other clouds to Amazon RDS for Oracle and reap the benefits of managed DR and reader farm capabilities.

By moving to a managed database service model, you can take advantage of managed high availability, DR, scalability, and a plethora of other capabilities that Amazon RDS for Oracle provides. You can now focus on what matters to your organization's business, such as optimizing database design, capacity sizing, and performance tuning. For more information on Amazon RDS for Oracle Read Replicas, see [Working with Oracle Read Replicas for Amazon RDS](#) and [Working with RDS Read Replicas](#).

About the Authors



Sameer Malik is a **Principal Database Solutions Architect** with Amazon Web Services.



Srinagesh Battula is a Principal Product Manager with Amazon Web Services.

TAGS: [Amazon RDS](#), [disaster recovery](#), [Oracle](#), [Oracle Active Data Guard](#), [RDS for Oracle](#), [reader farm](#)

Comments

0 Comments

Log

G

Start the discussion...